Dongwoo Kwak

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Experience

Dnotitia Seoul, South Korea
Optimization Engineer, AI Group – Inference Optimization *Feb 2026 – Present*

- Cut time-to-first-token 96% (24.4 s → 0.9 s) for a 5.7K-token agent prompt on an NPU-served Qwen3.6-35B-A3B by designing a session warm-up that pre-fills a byte-identical prompt prefix.
- Fit a 102 GB n-gram embedding table into 13.5 GB of GPU memory by porting 2-bit lookup-table quantization into vLLM with a fused Triton gather-decode kernel, 1.5× faster than a bf16 gather and bit-exact.
- Built a bilingual evaluation harness for a full tool-using agent stack, ranked 8 models on pass@1, and traced garbled-token failures to vLLM's default temperature 1.0 after ruling out tokenizer, GPU and parser.
- Benchmarked 3 serving engines incl. vLLM and SGLang for an in-house 2B embedding model behind a cosine ≥ 0.999 gate (66,810 tokens/s; 358 img/s, 3.1× SGLang) and exposed silent API bugs that cut MTEB 0.848 → 0.718.

Klleon Seoul, South Korea
AI Engineer, Product AI Team *Mar 2025 – Jan 2026*

- Fixed a GPU memory leak unresolved for 8 months by tracing missing shared-memory unregistration in Triton client teardown, reaching zero OOM crashes and retiring a forced restart every 3 days.
- Cut TTS request volume 75% and end-to-end latency up to 80% by building dynamic sentence batching on a CUDA-Graph GPT-SoVITS server inside the real-time digital-human engine (C++17).
- Reduced AWS GPU nodes 40% by moving media-engine processes off Triton inference nodes (10:1 → 50:3 ratio).
- Kept a lightweight face model in production by detecting broken frames at 0.85 ms/frame with masked histogram differencing (OpenCV, CUDA), doubling processes per pod.

Tmax Cloud Seongnam, South Korea
Researcher, Middleware Research Division *Feb 2024 – Oct 2024*

- Built a Triton-based model-serving IaaS (v0.0.0–v0.2.0) in C++20, porting the Triton core C API for asynchronous inference, health and statistics calls and adding tensor binary encoding.
- Sped up JSON writes 7× and reads 3× by replacing boost::property_tree with glaze in the middleware framework.
- Automated a 4-stage GitLab CI/CD pipeline (build, image, E2E test, deploy) with a Python E2E client.

Projects

vLLM Serving Toolkit | *Bash, Python, FastAPI, vLLM* 2026

- Wrote a layered-flag vLLM launcher, logging proxy and smoke harness measuring TTFT, TPOT and tool-call success; compared 7 models on RTX 6000 Ada vs GB10 and reused it in 4 later benchmark campaigns.

IRC Chat Server | *C++, Linux epoll* 2023 – 2024

- Migrated a multi-client IRC server from poll to epoll with EPOLLOUT toggling; added KICK and TOPIC commands.

Skills

Languages: C++17/20, Python, CUDA (working), TypeScript
Inference: vLLM, SGLang, NVIDIA Triton Inference Server, PyTorch/LibTorch, OpenCV, NPP, prefix/KV caching
Systems: Linux (epoll, sockets), Docker, Kubernetes, GitLab CI/CD, CMake, Meson, Conan, Grafana

Education

Yonsei University Seoul, South Korea
B.A. in Economics, Minor in Computer Science; GPA 3.56/4.5 *Mar 2018 – Feb 2024*

- Coursework: Operating Systems (ranked 10/130), Databases, Computer Architecture, Data Mining, Statistics.
- 42 Seoul software engineering program (2021 – 2023). Certifications: SQLD. English: OPIc IH.